

SYNOPSIS

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PROJECT TITLE:

Hospital Management Analysis Dashboard Using Power BI and MySQL

1. INTRODUCTION

The **Hospital Management Analysis Dashboard** is a Business Intelligence project developed using Microsoft Power BI and MySQL to analyse and visualize hospital data. The dashboard provides a centralized platform to monitor patient, doctor, pharmacy, laboratory, billing, and administrative information through interactive reports and KPI dashboards.

The project uses Microsoft Excel for data preparation, MySQL Workbench for database management, Power BI for data visualization, and Figma for designing the dashboard interface. The dashboard helps users analyse hospital operations efficiently and supports better data-driven decision-making.

2. PROBLEM STATEMENT

Hospitals generate a large amount of data across different departments. Analysing this information manually is time-consuming and makes it difficult to monitor hospital performance effectively. The lack of a centralized reporting system also affects quick decision-making. Therefore, an interactive dashboard is required to present hospital data in a simple, meaningful, and visually appealing format.

3. OBJECTIVES

1. To develop an interactive Hospital Management Dashboard using Power BI.
2. To store and manage hospital data using MySQL.
3. To analyse patient, doctor, pharmacy, laboratory, billing, and administrative information.

4. To create interactive KPI cards, charts, tables, and slicers for data visualization.
5. To provide a centralized platform for monitoring hospital operations.
6. To support better decision-making through visual analytics.

4. SCOPE OF THE PROJECT

The project focuses on analysing hospital-related data using interactive dashboards. It includes patient records, doctor details, pharmacy information, laboratory reports, billing details, and administrative data. Users can navigate between multiple dashboard pages, apply filters, reset selections, refresh data, and view reports through interactive visualizations.

5. TOOLS AND TECHNOLOGIES USED

Tool / Technology	Purpose
Microsoft Excel	Used for preparing, cleaning, formatting, and organizing hospital datasets before importing them into the database.
CSV Files	Used to store and transfer the prepared datasets between different platforms.
MySQL Workbench	Used to create, manage, and organize the hospital database by importing CSV files into database tables.
Microsoft Power BI Desktop	Used to develop interactive dashboards, reports, KPI cards, charts, tables, slicers, and visualizations
Power Query	Used for data transformation, filtering, data type conversion, and preparing data for analysis and reporting.
DAX (Data Analysis Expressions)	Used to create calculated columns, measures, KPIs, and analytical insights.
Figma	Used to design the dashboard user interface, navigation layout, and custom dashboard backgrounds

6. METHODOLOGY

The project follows the methodology outlined below:

1. Data Preparation:

- Hospital-related datasets were prepared and organized using Microsoft Excel.
- The datasets were cleaned, formatted, and saved in CSV format.

2. Database Management:

- The prepared CSV files were imported into MySQL Workbench.
- Database tables were created and organized to store hospital information.

3. Data Connection:

- Power BI Desktop was connected to the MySQL database.
- The required hospital data was imported for analysis and reporting.

4. Data Transformation:

- Power Query was used to clean, transform, and prepare the imported data.
- Data relationships and data models were created for effective analysis.

5. Dashboard Development:

- Interactive dashboards were created using Microsoft Power BI.
- KPI cards, charts, tables, slicers, and DAX measures were developed.
- Dashboard layouts were designed using Figma and applied as custom backgrounds in Power BI.
- Navigation buttons, Refresh button, Dark/Light theme, Notification page, and Clear Selection (Reset Filters) functionality were implemented to improve user experience.

6. Data Analysis:

- Patient, doctor, pharmacy, laboratory, billing, and administrative data were analysed.
- Interactive visualizations were used to generate insights and support data-driven decision-making.

7. DASHBOARD FEATURES

The Power BI dashboard includes the following key features:

1. **Home Dashboard** – Provides an overall summary of hospital operations through KPI cards and visual reports.
 2. **Patient Dashboard** – Displays patient information, admission details, discharge records, and patient-related analysis.
 3. **Doctor Dashboard** – Presents doctor details, department-wise information, and performance analysis.
 4. **Pharmacy Dashboard** – Provides medicine inventory and pharmacy-related insights.
 5. **Laboratory Dashboard** – Displays laboratory test details and related analysis.
 6. **Billing Dashboard** – Presents billing information, payment summaries, and financial analysis.
 7. **Hospital Management Dashboard** – Provides a consolidated view of major hospital activities and performance.
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8. EXPECTED OUTCOMES

- Provide a centralized view of hospital operations.
 - Simplify hospital data analysis.
 - Improve accessibility through interactive dashboards.
 - Support data-driven decision-making.
 - Identify operational trends and performance.
 - Reduce manual reporting efforts.
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9. PROJECT ADVANTAGES

- User-friendly and interactive dashboard.
 - Centralized hospital reporting.
 - Quick and easy data analysis.
 - Interactive filtering and navigation.
 - Professional dashboard interface.
 - Improves operational monitoring and decision-making.
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10. LIMITATIONS

- Uses a prepared dataset instead of live hospital data.

- Depends on the quality of the available data.
- Does not include predictive analytics.
- Not integrated with a live Hospital Information System (HIS).
- Focuses on analytics rather than complete hospital management

11. FUTURE ENHANCEMENTS

- Integrate with a real-time Hospital Management System (HMS).
- Implement Role-Level Security (RLS) for secure user access.
- Add predictive analytics for patient admissions and resource planning.
- Deploy the dashboard using Power BI Service for cloud access.
- Develop a web and mobile version of the dashboard.
- Integrate Electronic Health Records (EHR) for comprehensive hospital analytics.

12. CONCLUSION

The Hospital Management Analysis Dashboard successfully demonstrates the application of Business Intelligence techniques for analysing and visualising hospital-related data. The project integrates Microsoft Excel, MySQL Workbench, Power BI Desktop, Power Query, DAX, and Figma to transform raw hospital information into meaningful visual insights through interactive dashboards and reports.

The developed dashboards enable users to monitor patient information, doctor performance, pharmacy activities, laboratory tests, billing details, and administrative operations using interactive visualisations and KPIs. Overall, the project provides an efficient, user-friendly, and data-driven solution that supports informed decision-making and establishes a strong foundation for future development of a real-time hospital analytics system.